SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0502	**Course Title:**	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	**Assessment:**	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	**Total Marks:**	50 (10 Questions × 5 Marks)
CO4 Statement:	*Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)		
Student Name:	GUNALAN V	**Reg. No.:**	192325038
Section / Batch:	AI & ML	**Date:**	

Code of Conduct

I GUNALAN.V / 192325038 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________________

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

	Focus	Q.Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
	GRAND TOTAL:		50 Marks

Annexure A – Short Answer Test

1. How does Computer Vision enable computers to interpret and understand visual information?

a) What are the main objectives of Computer Vision?

b) Mention any two real-world applications of Computer Vision.

2. What are the key features of Computer Vision that make it suitable for intelligent image analysis?

a) Define feature extraction.

b) How does object recognition improve Computer Vision systems?

3. How are the different levels of image processing used in Computer Vision?

a) What is low-level image processing?

b) Differentiate between mid-level and high-level image processing.

4. How is a digital image represented and what are its fundamental components?

a) Define a pixel.

b) How does image resolution affect image quality?

5. How does image sensing and acquisition convert a real-world scene into a digital image?

a) What is the role of an image sensor?

b) Name any two image acquisition devices.

6. How do sampling and quantization contribute to the formation of a digital image?

a) Define sampling.

b) What is the purpose of quantization in image processing?

7. How does the image formation model describe the process of capturing images?

a) What is perspective projection?

b) What is the role of illumination in image formation?

8. How are image processing and Computer Vision related, and how do they differ?

a) Define image processing.

b) State two differences between image processing and Computer Vision.

9. How is Computer Vision applied in solving real-world problems across different domains?

a) Explain one application in healthcare.

b) Explain one application in autonomous vehicles.

10. How do digital image fundamentals support Computer Vision applications?

a) Why are pixels considered the basic building blocks of digital images?

b) How do sampling and image resolution influence image analysis?

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: ______________	Date: ______________	Date: ______________

1. Answer:

Computer vision enables computers to acquire, process, analyze, and interpret images or videos so they can identify objects, patterns and scenes like humans

a.) Main objectives of computer vision:

* Image understanding
* Object detection and recognition
* Image classification
* scene analysis
* Decision-making based on visual data

b.) Two real-world applications:

* Facial recognition
* self-driving cars.

2. Answers:

Computer vision uses feature extraction, object recognition, image segmentation, pattern recognition, and machine learning to analyze image intelligently.

a.) Define feature extraction

Its a process of identifying important characteristics (edges, corners, textures, shapes) from an image.

b.> How does object recognition improve computer vision?

It idefies and classifies objects accura enabling automated decision making.

3. Answer :

i> low-level : Image enhancement and noise removal.

ii> Mid-level : segmentation and features extraction

iii> High-level : Object recognition and scene understanding.

a.> what is low-level image processing?

It improves Image quality by reducing noise, sharpening, and enhancing contrast.

b.> Differentiate between mid-level and high-level image processing.

Mid-level	High-level.
segmentation and feature extraction	Object recognition & scene understanding
Produce image features	Produces meaningful interpretation

Answer:

A digital image is represented as a matrix of pixels. its components include pixels, resolution, brightness, and colour values.

a.) Define a pixel

A pixel is the smallest unit of a digital image containing intensity or color information.

b.) How does image resolution affect image quality?

Higher resolution provides more detail and better image quality.

5. Answer:

Light reflected from objects is captured by an image sensor, converted into electrical signal, and digitized into an image

a.) What is the role of an image sensor?

It converts incoming light into electrical signals.

b.) Name any two image acquistion devices

* Digital camera
* scanner.

6. Answer:

Sampling converts a continuous ima[illegible] into discrete pixels, while quantization assigns intensity values to each pixel.

a.) Define sampling

Sampling is the process of converting a continuous image into discrete pixels.

b.) what is the purpose of quantization?

It assigns finite intensity values to pixels for digital representation.

7. Answer:

The image formation model explai[illegible] how light from a scene passes through a camera lens and forms an image on the sensor.

a.) what is perspective projection?

perspection projection maps a 3D scene onto a 2D image while preserving depth.

b.) what is the role illumination in ima[illegible] formation?

Illumination provides light necessary for capturing clear and detailed images

Image processing enhances images whereas computer vision interprets and understands them.

a.) Define image processing.

Image processing is the technique of enhancing or modifying digital images.

b.) Two differences:

1. Image processing improves image quality; computer vision extracts meaningful information.

2. Image processing is image-focused; computer vision is decision-focused.

9. Answers

computer vision automates image analysis in healthcare, transportation, agriculture, surveillance and industry.

a.) Healthcare application:

Detecting diseases from x-rays, MRI or CT scans.

b) Autonomous vehicle application:

Detecting roads, traffic signs, pedestrians and obstacles for safe driving

10. Answers

Digital image fundamentals provide accurate image representation for analysis, recognition, and decision-making.

a) why are pixels considered the basic building blocks of digitals images?

Because every digital image is made up of pixels, each storing brightness or color information.

b) How do sampling and image resolution influence image analysis?

Proper sampling and higher resoluti preserve image details, leading to more accurate feature extraction and object recognition.